Equipment

ONR FY2027 Young Investigator Program, NOFO N0001426SF004. PI: Sterling G. Baird, Brigham Young University (BYU)

Equipment available. The equipment below is in place and available to the project at no cost, except where a facility charges hourly rates, which are in the budget.

- AMAZEMET rePOWDER ultrasonic atomizer with an induction crucible module (to 1,400 °C, typically run to 1,300 °C; Facilities, Fig. 1), with tungsten-coated carbon-fiber plates, boron-nitride-coated graphite crucibles, and ceramic or ceramic-coated sealing rods and nozzles wherever the copper-alloy melt touches (PI laboratory, BYU; purchased with start-up funds in 2026).
- Liquid-handling gantry (CubXL, Ursa Laboratories; Fig. 1b) with a pipette, a vial capper and two cameras, run by our own scripts (PI laboratory). It runs the electrochemical corrosion screen and samples the immersion solutions.
- Powder handling, sieving and specimen preparation equipment in the PI laboratory: an open-source auger powder doser built in the laboratory (Fig. 1a), a sieve stack from 45 to 250 µm with a sieve shaker, a Hall flowmeter funnel, an analytical balance, glove bags, dry powder storage and sectioning and mounting equipment. The atomizer's steel-panelled enclosure (exhaust, argon supply, dehumidifier, strip-curtain door) with an ATEX-rated vacuum cleaner, a Class D extinguisher and full-face respirators.
- NuVant EZstat Pro potentiostat (direct-current measurements), lent to the PI laboratory by a BYU colleague and run by our own scripts on the gantry; it has run the electrochemical corrosion screen since October 2026 and does so until the requested instrument arrives (Fig. 2).
- Scanning electron microscopes with EDS and EBSD (Apreo C, Helios 600 with a focused ion beam, Verios) and a Tecnai TF-20 transmission electron microscope at the BYU Electron Microscopy Facility, at the posted rate of $30 per hour for engineering users.
- Agilent 8900 ICP-MS/MS and XPS at the BYU Spectroscopic Facility, and X-ray diffraction at the BYU X-ray Diffraction Facility.
- Metallographic polishers (PX500 automatic, PX300 manual and vibratory), a Keyence VHX-7000 digital optical microscope, and CNC machining at BYU.
- Instron load frames in the materials testing laboratory of the Department of Mechanical Engineering, for sub-size tensile tests.
- The BYU Office of Research Computing cluster (about 28,000 CPU cores and 367 GPUs).
- Blown-powder DED through purchased sessions: the InssTek MX-Lab (500 W, six powder feeders) at the University of Utah for tracks and small deposits from up to 0.5 kg of powder, and the FormAlloy X2 at the Colorado School of Mines for 2 kg builds and repair coupons.
- Gas atomization through one purchased run a year at Ames National Laboratory (Dr. Jordan A. Tiarks, unfunded collaborator), which turns the year's discovered alloy into a gas-atomized lot.



(a) (b)

Figure 1. Equipment in the PI laboratory. (a) The open-source auger powder doser over its analytical balance in a fume hood, dosing an atomizer charge (September 2026). (b) The CubXL liquid-handling gantry with its vial rack and pipette. Frames from the laboratory's videos.

Equipment to be purchased with project funds. One item, in Period 1: a research-grade single-channel potentiostat with impedance capability, with corrosion and impedance software, a flat cell, a Faraday cage and reference electrodes (such as a Gamry Interface 1010E or a BioLogic SP-150e), $18,000, a budgetary estimate from current list prices. The electrochemical corrosion screen is one step of the workflow that the cost-aware search schedules, so the instrument must be available every day under our own scripts on the gantry, and the screen's impedance measurements need a capability the loaned instrument lacks. The potentiostats at BYU are shared instruments booked by the hour, which the search cannot schedule. The item is excluded from the MTDC base (Budget Justification, section D).

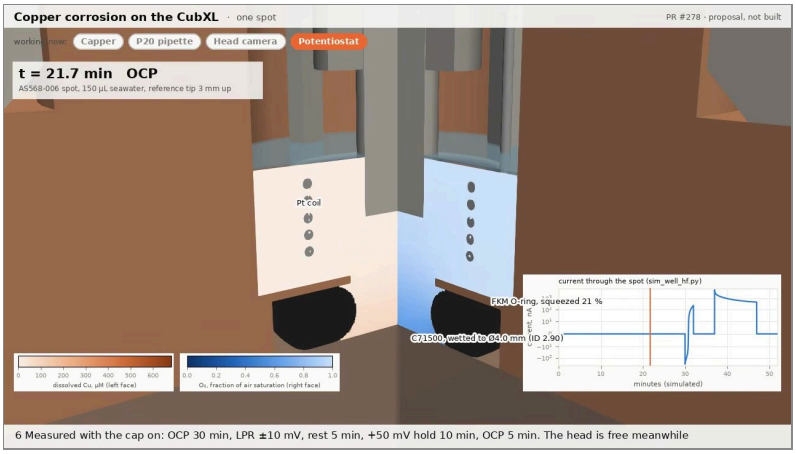


Figure 2. The copper corrosion screen on the gantry, from the laboratory's animation of the planned protocol: an O-ring cell on a copper-nickel specimen during the open-circuit potential step, with the simulated current through the spot. The requested potentiostat will drive this step; the loaned instrument listed above runs it now.

Government-owned equipment. None.